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Bioinformatics

- **bioinformatics:**
  - → is conceptualizing biology in terms of molecules
  - → applying informatics techniques to understand and organize the information associated with these molecules, on a large scale

⇒ Bioinformatics is a management information system for molecular biology and has many practical applications

- **Biological information:**
  - → Bio + Informatics: "information science of the biological world"
  - → DNA, RNA, Protein: information bundles of the cell
  - → levels: 1) Sequence and Structure
    2) Gene / protein expression, DNA mutations
    3) Network of interactions among the macromolecules

- **Databases:** Protein Sequence, Nucleotide Sequence, Macromolecular structures, Genome Sequence, Protein-Protein interactions, Transcription Factor Binding Sites

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Biological Basics

Cells

Prokaryotic Cells : eg. Bakterien
Eukaryotic Cells : eg. Potato

Organization of the Genome

- Size of eukaryotic animal cell 10 - 30 µm
- Size of nucleus : 6 µm
- Length of DNA : 3 m

Central Dogma of Molecular Biology

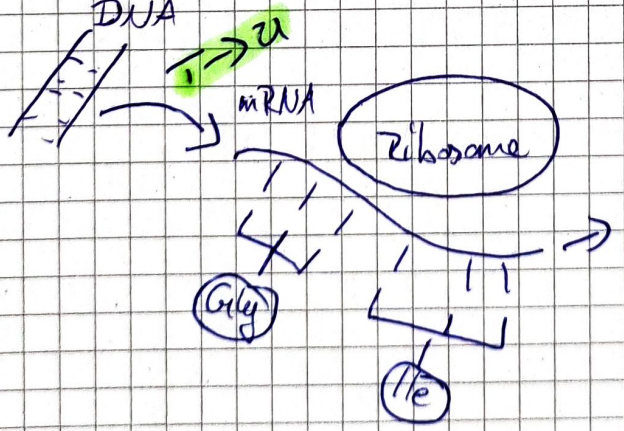
DNA ═══ "the information content of the cell"

↓ Transcription

m RNA ——— "flow of information in the cell"

↓ Translation

Protein ⬭ "functional unit of the cell"



Codon : Triplet of nucleotides that code for a specific amino acid

AUU
AUC → Ile

GGU
GGC → Gly
GGA
GGG

Start : AUG

Stopp : UAA, UAG, UGA

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Genome

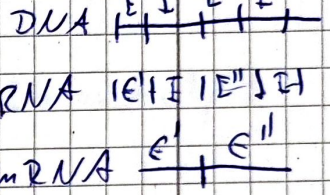
Human : 46 Chromosomes
with 20 - 25 K protein coding genes

→ 98% of the genome is non-coding DNA
2% is coding
7-9% functional
→ in bacteria 2% are non coding

Introns and Extrons

DNA is interrupted by short seq's that are not in the final mRNA.

↳ introns
↳ Extrons = RNA kept in the final seq



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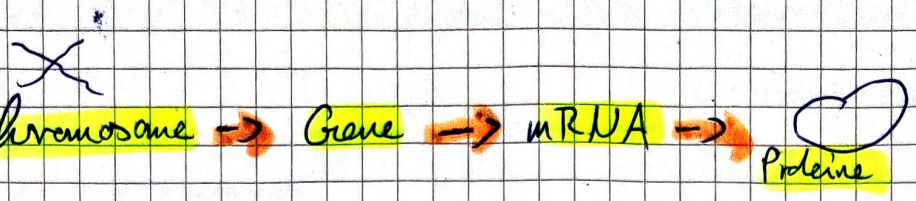
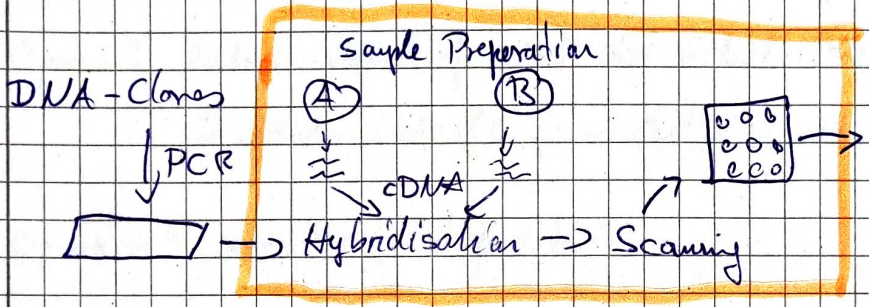
Gene Expression



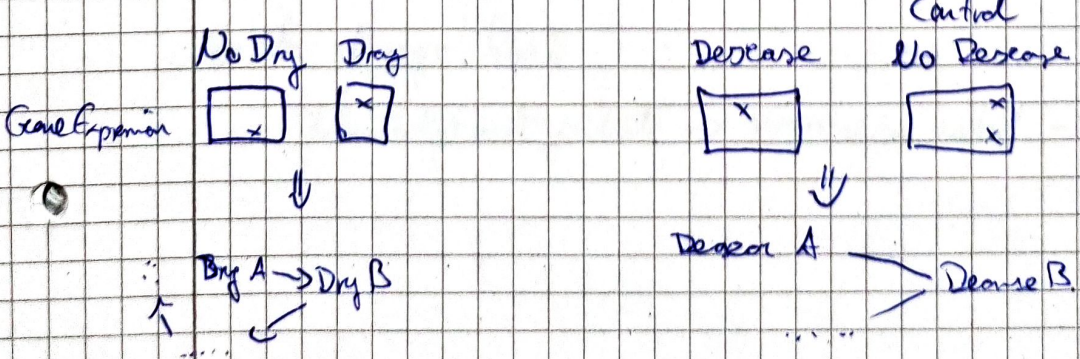
- Gene: "Segment of DNA that codes for a specific protein"
- Gene Expression: "activation of a gene to synthesize a protein"
- Gene Expression Array:
  a miniature, massively parallel and semi-automated process that measures levels of gene products



- Gene Expression Analysis:

RAW Data ⇒ Preprocessing ⇒ Statistical testing ⇒ Functional Analysis ⇒ Biological interpretation

Comparative Gene Expression Analysis:
The study of differences in gene expression patterns
eg: identify a mutation that causes a disease



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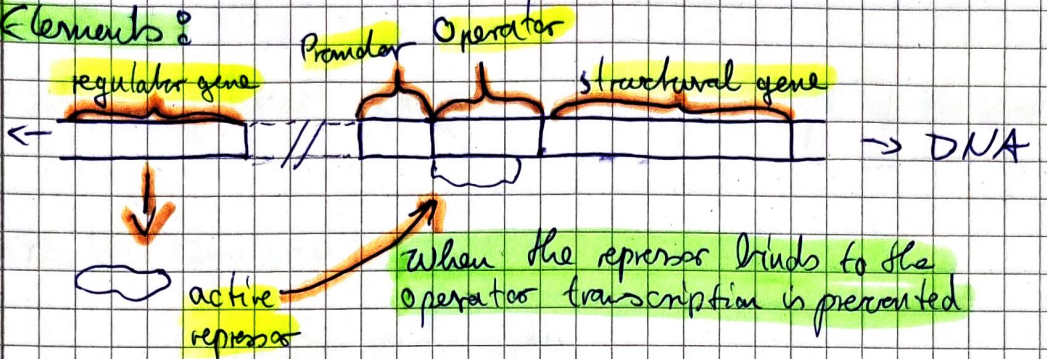
Gene Regulation

- Same DNA Yet Different functions → Diff in Gene Expression
  ↳ Gene Regulation

Prokaryotes

Elements:



- Operon: entire stretch of DNA that includes the operator, the promoter and the genes that they control
- Operator: repressor binding site
- promotor: site of transcriptional activation
- Inductible operon: operon is activated by small molecule called inducers, eg: lac
- Repressible Operon: operon is shut-down by small-molecule called co-repressor, eg: trp

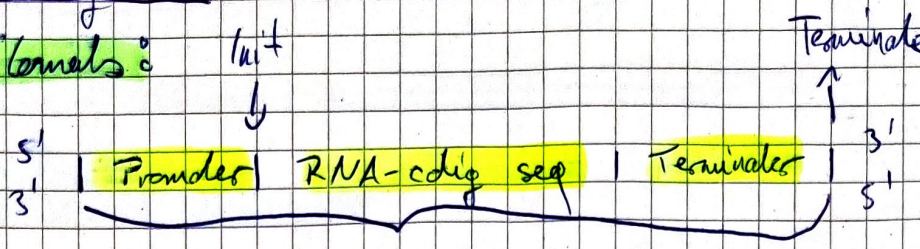
Levels of regulation:

Transcriptional control → Translation control → Post-Translation control

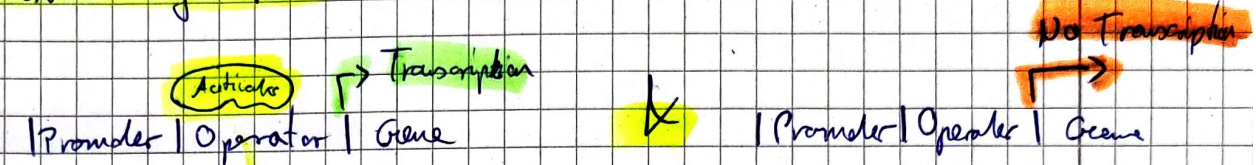
GBB

Eukaryotes

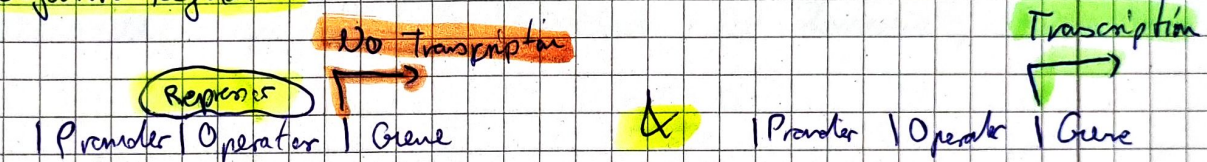
Elements:



- Gene Regulation: the process of turning on/off of the genes
- Positive Regulation:



- Negative Regulation:



- Enhancers: can be bound by proteins to increase transcription rate of a gene (Seq's in DNA).

- Transcriptional Control:
  a biological process that allows the cell or an organism to respond to a variety of intra- and extra-cellular signals,

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Reading the Genetic Code DNA Sequencing

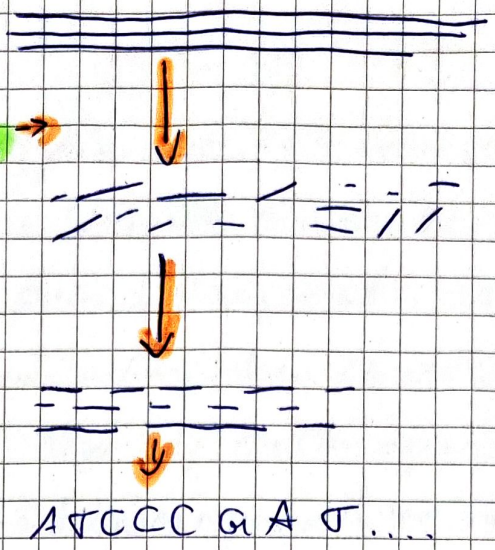
Cloned Genome →

Multiple Genome are sheared into small pcs →

unordered segments →

Assembly →

Overlaps →



Genome Length

Human , 3 billion Basepairs , 46 Chromosomes , ~25K Genes

E. coli , 4.6 million , 1 Chromosome , 3200 Genes

Human Genome:

Ø Size Chromosome: ~130 Mega ~~Bytes~~ bps

Genes per chromosome: ~300

Ø size of gene: 1148 - 37,7 Kilo bp

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Comparing DNA Sequences

Sequence alignment

Pairwise Seq Alignment: exactly two seq's are aligned to each other to determine the similarity between the seq's

Multiple Seq Alignment: process of aligning more than two seq's to identify similarities between the seq's

Local Alignment: will align only similar regions between seq's, ~~but~~ and leaves region with too many diff unaligned

Global Alignment: Attempt to align every base in each aligned seq

- Gap: Space introduced into an alignment to compensate for In/Dels in one seq relative to another.
- Gap penalty: when a seq alignment is scored, gaps in the alignment are given a negative score or gap penalty

BLAST = Basic Local Alignment Search Tool

↳ Sequence comparison tool

↳ used to search in sequence databases for optimal local alignments (to a query)

↳ The alignments are assigned an significance score i.e. E-value

E-value: The expectation value is the number of different alignments with scores equivalent to or better than a score S that one can expect to occur in a database search by chance. The lower ~~time~~ the E-value, the more significant the score S, thus, the alignment.

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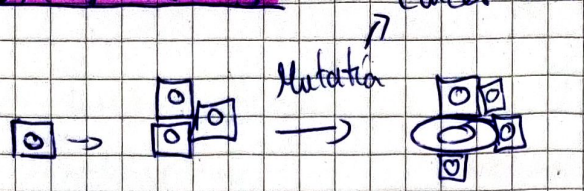
DNA-Mutations



-> Deletion, Duplication, Inversion, Insertion, Translocation

Unraveling Gene Regulation

ChIP-Seq procedure: Experimental detection of Protein-DNA binding events

↳ sequencing of the genomic DNA fragments that co-precipitate with a DNA-binding protein that is under study.
Technik um Bindungsstellen von TF's auf der DNA zu detektieren.
Protein-DNA-Interactions

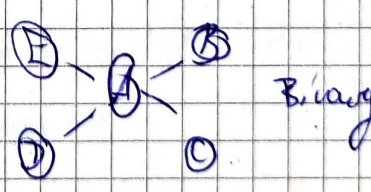
↳ Prediction of TFBS!

=> key tech for epigenetics

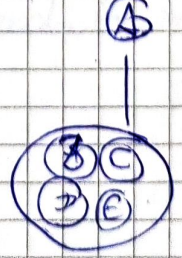
-> detect DNA binding protein, HM to identify enriched loci

Interaction Networks:

Diff Kinds



Binary



Co-complex

GSB - 01

1) Basics of molecular biology

Eukaryotes	Prokaryotes
-> Nucleus	-> No Nucleus
-> 10 - 100 µm	-> 0,1 - 5 µm
-> mostly multicellular	-> unicellular
-> linear DNA	-> circular DNA
-> Chromosomes	-> No Chromosomes

2) Central Dogma of molecular biology

The flow of information in the cell consists of 2 steps.

First the DNA is transcripted to mRNA and second the mRNA is translated to a protein

3) Comparative Gene Expression Analysis?

The study of differences in the cell gene expression patterns.

Eg: identify a mutation that causes a disease.

4) Gene Regulation

Gene regulation is the process used to control the timing, location and amount in which genes are expressed.

Positive Regulation: activator binds to operator to enable transcription

Negative Regulation: repressor binds to operator to disable transcription

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A - T (U)
C - G

2) Exon intron Exon

5' ...CAAGAT... G gtaag...g GAT... 3'

a) → mRNA = CAAG AU...G GAU...

b) Amino Acid Seq:

Start: AUG — Stop: ACG

Met, ..., Stop

Immer Drei übersetzen!

Mutation:

In promotor: increase / decrease of gene expression.

In Exon: could be change in Protein Seq

In Intron: No change

DNA Synthesis: Occur during DNA Replication!

DNA Transcription to RNA: In promotor more or less ~~Exon~~ gene transcription!
↳ changes binding sites for TF...

DNA Translation: by Mutation in Exon!

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Genomics

→ Study of whole genomes of organisms & incorporates elements from genetics

→ uses a combination of recombinant DNA, DNA Seq. Methods & bioinformatics to sequence, assemble and analyse the structure and function of genomes

→ Genomics : addresses all genes and their interrelationships to identify their combined influence on the growth and development of the organism. eg interaction between Alleles & loci

→ Genetics : ~~scr~~ scrutinizes the functioning and composition of the single gene ~~whole~~

→ Sequence whole genome : Fred Sanger & next generation sequencing

Sequencing

→ getting access to the seq or primary structure of any Biopolymer

Next Generation Sequencing

→ high throughput & highly parallel sequencing → Next Generation Sequencing

Sanger vs NGS

Sanger : 10 year Human Genome Project ; first human reference genome ; cost ~ 3 $ billion → for complex disorders and cancer

HiSeq : Today < 4K$ in less than 2 weeks on a single sequencing machine

↳ More accurate

GDB

Sequencing by Synthesis

→ Start from a single strand a DNA polymerase incorporates the four bases with different labels.

→ carried out in real time

→ eg: PacBio a real time single molecule sequencing technology

PacBio SMRT sequencing

→ Fluorescently labeled nucleotides!

↳ long reads, short sequencing time, no amplification bias, can detect modifications but but large instrument

Short vs long read tech

▷ Inherent limitations of short read technologies:

→ GC bias → difficulties mapping to repetitive elements

→ trouble discriminating paralogous seq's

→ difficulties in phasing alleles

▷ long read single molecule sequencers resolve obstacles:

→ higher consensus accuracies

→ can detect epigenetic modifications from DNA

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General Sequencing Bias

Common Source for bias:

-> Decreased coverage in regions of high or low GC.

↳ reason: PCR amplification during library construction is a known source of undercoverage of GC-extreme regions

-> Homopolymers (eg: AAAA)

-> Strand specific bias

-> Terminator-free chemistry may limit ability to accurately sequence long homopolymers

-> PacBio (lowest Bias)

-> Analysis Bias

Steps in NGS pipeline

DNA extraction ⇒ Genetic enrichment ⇒ Library prep ⇒ Sequencing ⇒ Primary analysis ⇒ Secondary Analysis

Problems:

Mapping: describes the process of finding the region where a read is placed in the context of a reference sequence.

Alignment: describes the exact positioning of each base in a read in a target region

Assembly: Merge a set of reads in order to reconstruct an unknown seq.

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Assembly

- DNA:
  - I) DNA - reconstruction of genomes
    Reconstruct the genomes of organisms where no reference sequence is available.
  - II) DNA - local reconstruction
    Only parts need to be assembled (In / Dels's)

- mRNA:
  - I) RNA - messenger RNA
  - II) RNA - micro RNA

Solution

-> Graph based
  -> nodes : seq fragments
  -> edges : by overlaps
  } Overlap graph

-> Greedy: Hamiltan, de Bruijn Graph

-> Hamiltan path:
  -> A path that visits each node exactly once
  -> Determine if such a path exists is NP-Hard

-> Eulerian path:
  -> A path that visits every edge exactly once
  -> Polynomial time algorithm exists

Cn 5B

Simple greedy algorithm

Greedy

I) Compute overlap between all fragments (suffix to prefix)

II) Find the heaviest edge and merge the nodes

III) Iterate till only one node is left

Problem:

Repetitions:

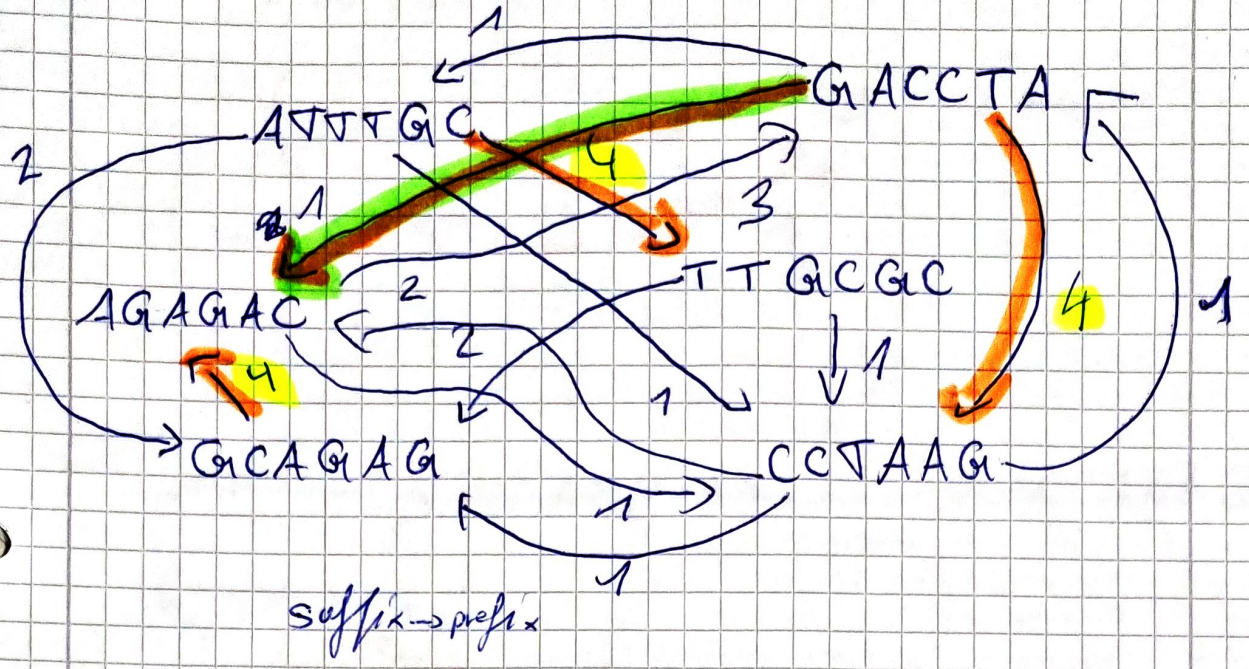
→ The greedy alg focus on local optimization which lead to problems in the assembly process

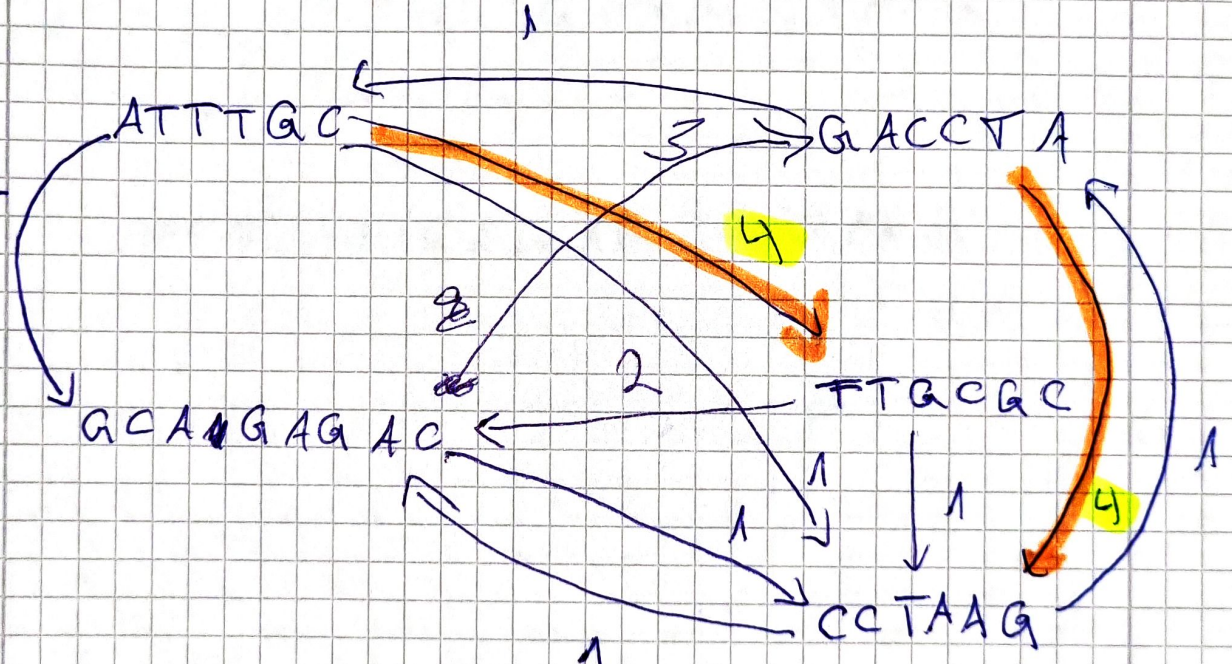
→ Reads may have increased coverage in NGS experiments

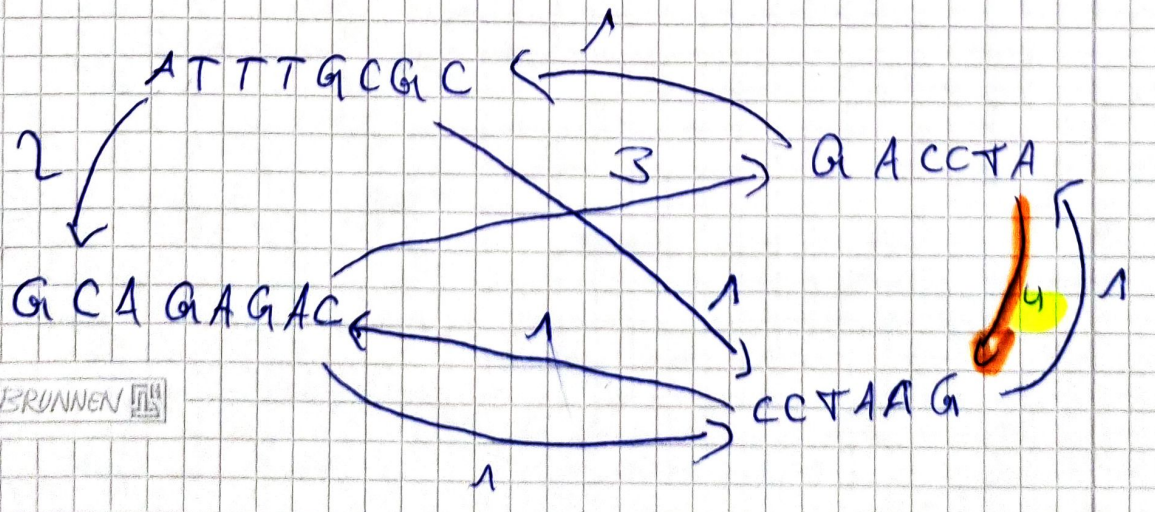
BRUNNEN

GSB - 6x2

1) Assembly Problem - Simple Greedy Alg







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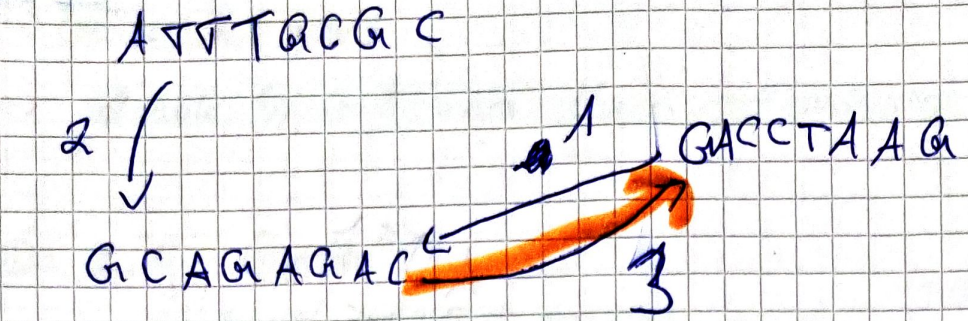
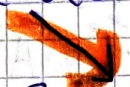
GSD - 62

1)



~~ATTTGCGCAGAGACGACCTAAG~~

ATTTGCGC



2 GCAGAGACCTAAG

ATTTGCGCAGAGACCTAAG

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Else Eulerian path

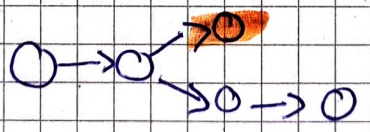
→ read k-mers

→ calculate for the k-mers the # of occurrences

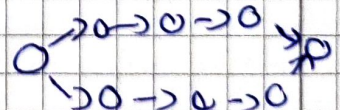
- De Bruijn Graph:
  - → Construction of the De Bruijn Graph (DG)
    n-dimensional graph on an alphabet with m symbols.
  - → Overlap Graph
  - → m^n nodes
  - → If a node results from another node by shifting the seq by one char to the left and concatenating a new char at the end, an edge is drawn (k-1-mers) ACG → CGA
  - → Eulerian ~~ACGA~~ ACGA ⇒

⇒ the assembly problem ⇔ finding an Eulerian path that traverse each edge in the de Bruijn graph

- Artifacts: I) Sequencing error at the end of a read → SPUR

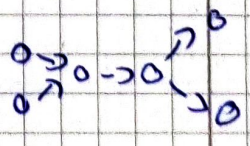


II) Paths that diverge & later converge → BUBBLE
  - ↳ actual variant / mutation
  - ↳ Sequencing error in the middle
  - ↳ frequently PCR artifacts



III) Cycles → Repeats

IV) Paths that converge & later diverge → frayed rope
  - ↳ indicate repeated elements



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Eulerian

(+) No explicit calculation of alignments → fast

(+) much less complex polynomial time!

(−) there is usually not a single Eulerian path

GSB – Ex 2

(2) De Bruijn Graph

a) 4-mers

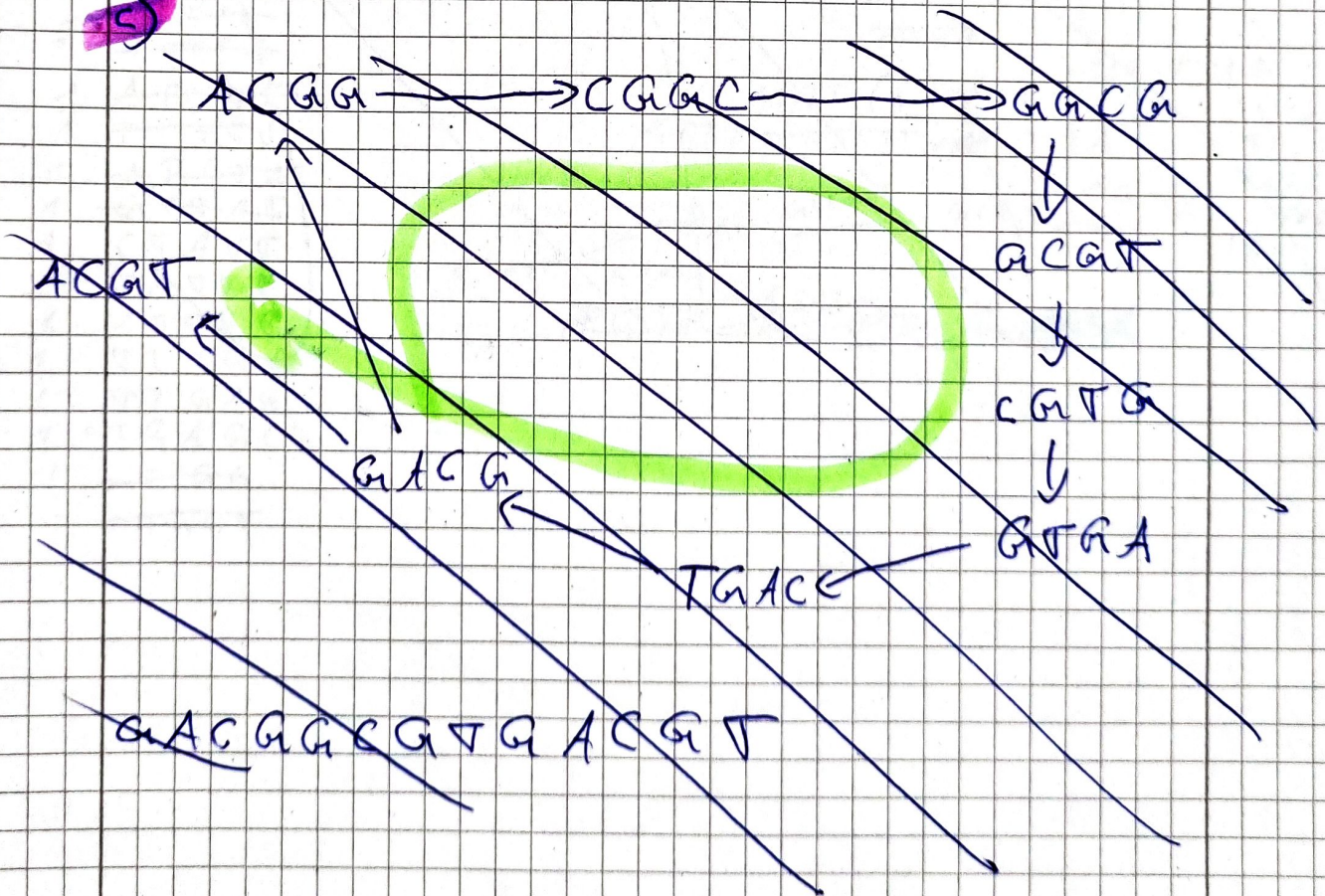
ACGGC : ACGG, CGGC ; CGGCG : CGGC, GGCG
CGTGA : CGTG, GTGA ; GACGT : GACG, ACGT
GCGTG : GCGT, CGTG ; GGCGT : GGCG, GCGT
GTGAC : GTGA, TGAC ; TGACG : TGAC, GACG

b)

Count of 4-mers :

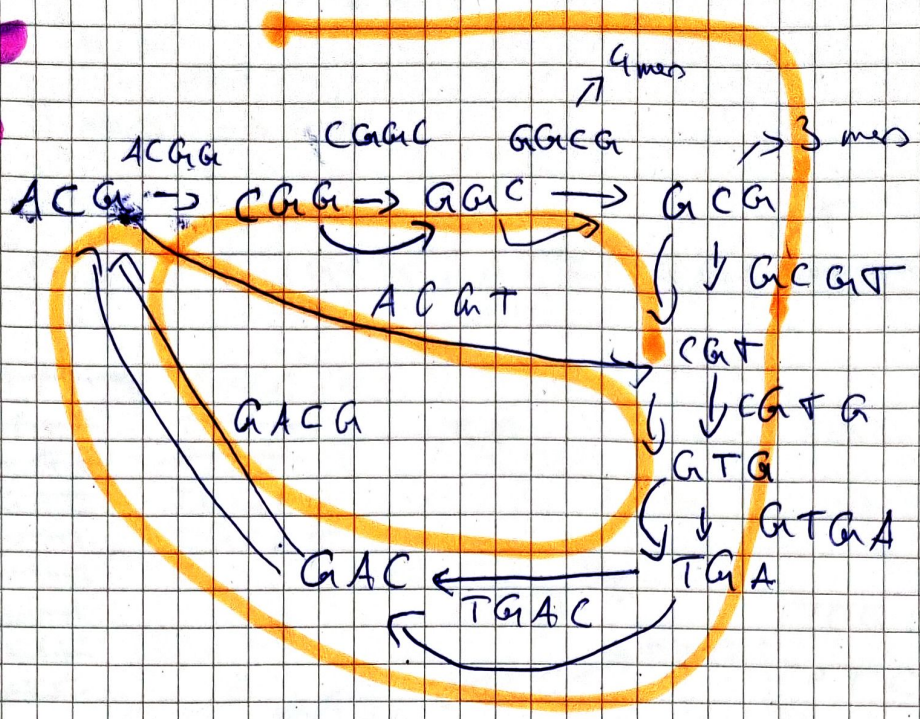
ACGG : 1 , CGTG : 2 ✓ , TGAC : 2
ACGT : 1 , GTGA : 2 ✓ , GACG : 2
CGGC : 2 ✓ , GCGT : 2 ✓ , GGCG : 2

c)

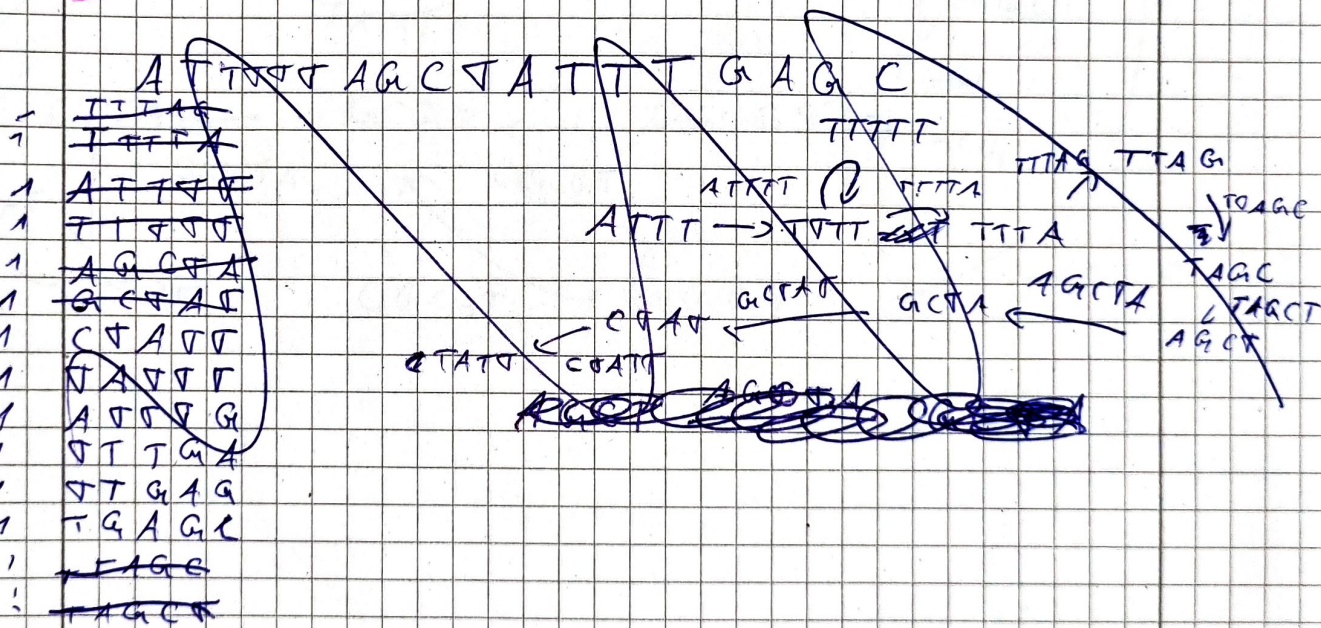


GSB 42

c)

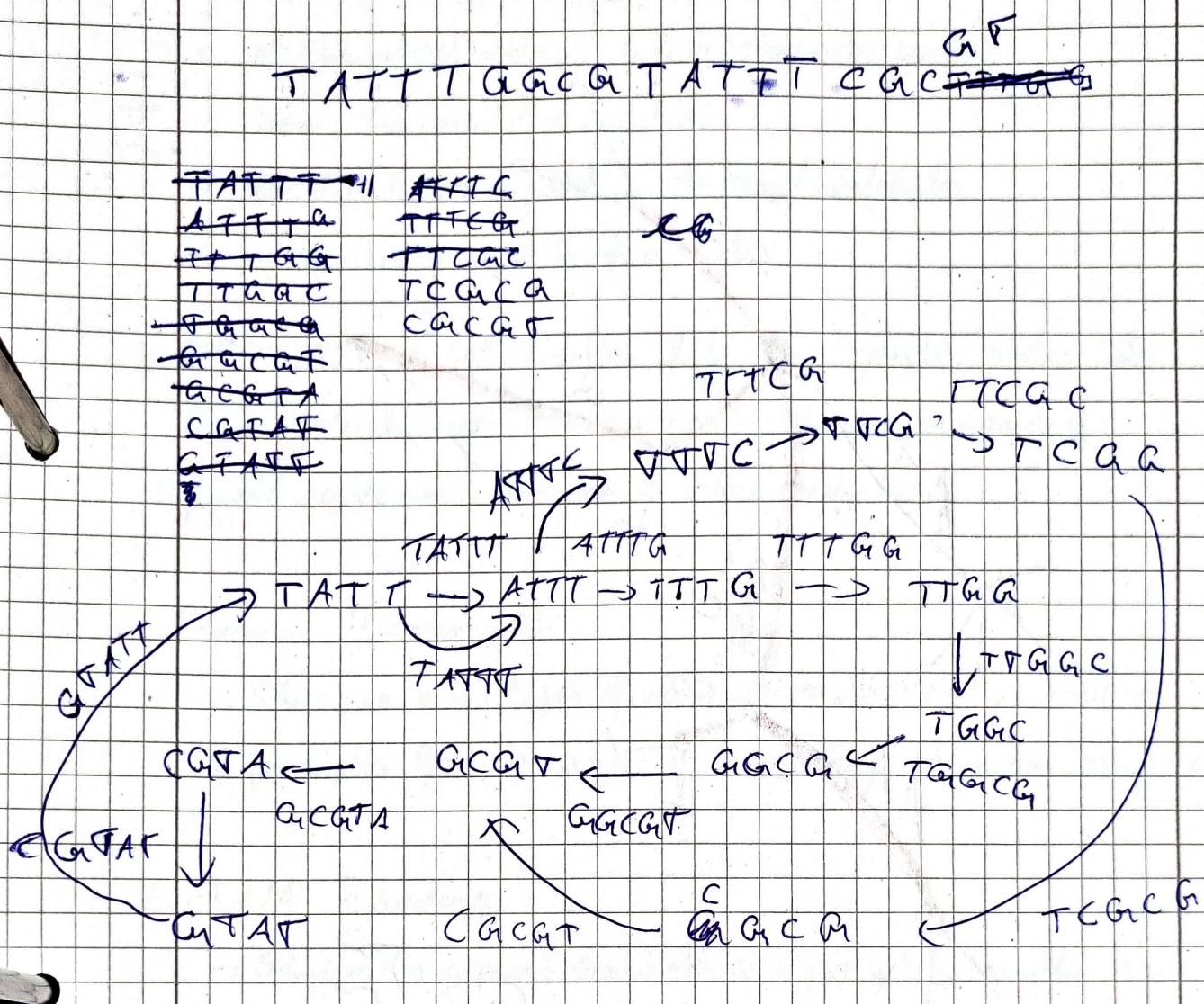


d) Bubble, 5-mers



GSS 16x2

b) 5 mers, Bubble



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Microarrays

-> experimental format based on the synthesis or attachment of probes, which represent genes
-> using a solid substrate
-> intended to be exposed to the target molecules
-> quantitative measure of mRNA

The level of hybridization between specific probes and target molecules which is generally indicated by means of fluorescence is measured by image analysis.

- **Expression Microarrays:**
  -> There are many types of microarrays. They rely on similar principles but the details of its operation change from one to another case

- **cDNA Technology**
  -> Samples are prepared from both an experimental sample and a control sample and are then overlaid on the array and allowed to hybridize to each spotted probe.

(I) -> control only expressed gene.
(II) -> experiment only expressed gene.
(I) & (II) -> gene expressed in both.
  ↳ Report ratio of gene expression

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Normalization

Data pre-processing : Background subtraction
Normalization
Collapsing / imputation

Why normalize data?

→ adjust the intensities from each array to balance them so that meaningful biological comparisons can be made.
→ unequal quantities of starting RNA
→ Differences in labeling and hybridization efficiency
→ Differences in detection efficiencies between the fluorescent dye used
→ Systematic bias

→ Without normalization, many genes will appear to be upregulated when they in reality are not.

Methods: Scaling, LOESS, RMA

Assumption: no global changes in gene expression between samples

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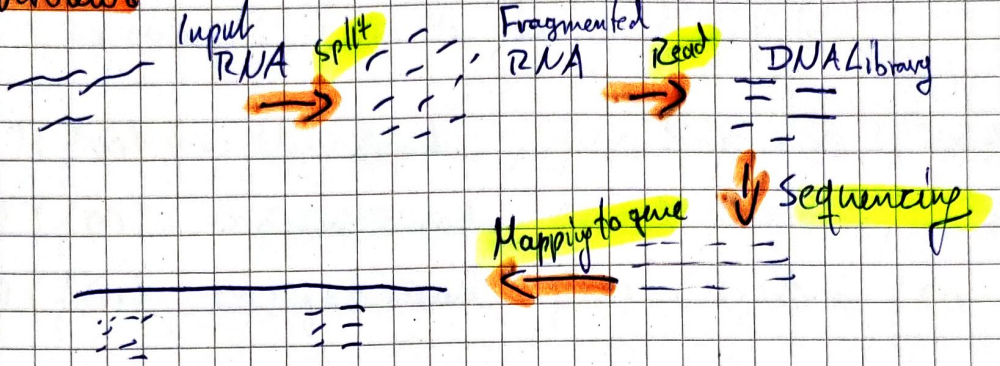
RNA-Seq

application : Transcript quantification,
Transcript discovery,
Expression Profiling

Overview :



Data Analysis :

Raw Data → QC → Spliced Read Alignment → Transcript Reconstruction
↓
Differential Expression analysis
↓
Biology

Count Matrix :

$N = n \times m$, N_{ij} = number of reads in gene i sequenced in experiment j

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Algorithms

An algorithm is a sequence of instructions that one must perform in order to solve a well formulated problem.

↳ correct : $\forall$ In to correct Out

↳ incorrect : $\exists$ In to incorrect Out

$O(f(x))$ ⇒ Upper Bound , $f(x) \leq c\, g(x)$, $x \geq x_0$

$\Omega(f(x))$ ⇒ Lower Bound , $f(x) \geq c\, g(x)$, $x \geq x_0$

$\Theta(f(x))$ ⇒ Tight Bound , $f(x) = O(g(x)) \;\&\; f(x) = \Omega(g(x))$

$$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$$

polynomial runtime $O(n^p)$

exponential runtime $O(p^n)$

Dynamic Programming :

→ Break Problems into subproblems and solve them

→ Keep track of repeating computation in Dynamic Programig Table

Divide and Conquer

→ divide into smaller subproblems and solve them then merge together

→ $O(n \log n)$

GdSB - 6x3

1) Time Complexity and Big O-Notation

a) $O(n \log n)$ algorithm

1 ms → 1000 data

$T(n) = c\, n \log n$

~~1 000 000~~

$T(1000) = c \cdot 1000 \cdot \log(1000) = 1$ ms

$\Leftrightarrow c = \frac{1}{3000}$ ms

$T(1\,000\,000) = \frac{1}{3000}$ ms $\cdot\ 1\,000\,000 \cdot \log(1\,000\,000) = 2000$ ms

b)

1) $5 + 0{,}001n^3 + 0{,}25n = O(n^3)$
2) $500n + 100n^{1,5} + 50 n \log_{10} n = O(n^{1,5})$
3) $0{,}3n + 5n^{1,5} + 2{,}5n^{1,75} = O(n^{1,75})$
4) $n^2 \log_2 n + n(\log_2 n)^2 = O(n^2 \log_2 n)$
5) $n \log_3 n + n \log_2 n = O(n \log n)$
6) $3 \log_8 n + \log_2 \log_2 \log_2 n = O(\log_8 n)$
7) $100n + 0{,}01n^2 = O(n^2)$
8) $0{,}01n + 100n^2 = O(n^2)$
9) $2n + n^{0,5} + 0{,}5n^{1,25} = O(n^{1,25})$
10) $0{,}01n \log_2 n + n(\log_2 n)^2 = O(n(\log_2 n)^2)$

Ü5 B

c)

- $\mathcal{O}(f+g) = \mathcal{O}(f \cdot g)$ ↯

  ↳ $\mathcal{O}(f+g) = \mathcal{O}(\max\{f, g\})$

- $\mathcal{O}(f \cdot g) = \mathcal{O}(f) \cdot \mathcal{O}(g)$ ✓

- $g = \mathcal{O}(f)$ & ~~$h = \mathcal{O}(f) \Rightarrow g = \mathcal{O}(h)$~~ ↯

  $f = \mathcal{O}(f)$ & $f = \mathcal{O}(h) \Rightarrow g = \mathcal{O}(h)$

d) up to 10^9 records

$T_A(n) = 0.001\,n$ ms

$T_B(n) = 500\sqrt{n}$ ms

$T_B(n) = \mathcal{O}(\sqrt{n}) < T_A(n) = \mathcal{O}(n)$ bei n_0 ?

$500\sqrt{n} \leq 0.001n \quad | : 0{,}001$

$500\,000\sqrt{n} \leq n$

$2{,}5 \cdot 10^{11} \leq n$ → bei only up to 10^9 records

↳ $T_A(n)$ is here better!

e)

```
for (i=1; i<n; i*=2){                    O(log2 n)
  for (j=n; j>0; j/=2){                  O(log2 n)
    for (k=j; k<n; k+=2) {               O(n)
      sum += (i+j*k);                    O(1)
    }
  }
}
```

$\mathcal{O}((\log_2 n)^2 \cdot n)$

GSB

(2)

a)

```
Majority (array) :
    if (len(array) == 1)
        return array[0]
    mid = len(array) // 2
    L = array[:mid]
    R = array[mid:]
    left_m = Majority(L)
    right_m = Majority(R)
    if left_m = right_m :
        return left_m
    l_cont = 0 ; r_cont = 0
    for in range (len(L)) :
        if L[i] == left_m :
            l_cont ++
    for i in raje(len(R)) :
        if R[i] == right_m :
            r_cont ++
    if l_cat > r_cont :
        return left_m
    else
        return right_m
```

GSB-Ex 3

b) Boyer-Moore Alg

```
BoyerMoore Alg (array , l)
    count = 0
    candidate = 0
    for (int i=0 ; i< l; i++) {
        if (count == 0)
            candidate = array[i]
        if( array[i] == candidate) {
            count++
        else
            count--
    return candidate
}
```

c) no Majority

↳ Last digit that was set as candidate